

ECON 636 600:MACROECONOMIC THEORY I

Homework 1 (due on 10:00am Tuesday, Sep 11)

1. (50 points) Derive the Euler equation

$$u'[c_t] = \beta u'[c_{t+1}] f'(k_{t+1}); t = 0, \dots, T-1$$

by applying the Kuhn-Tucker condition to the following problem,

$$\begin{aligned} \max \quad & \sum_{t=0}^T \beta^t u(c_t), \\ \text{s.t.} \quad & c_t + k_{t+1} \leq f(k_t), t = 0, \dots, T \\ & -k_{T+1} \leq 0. \end{aligned}$$

Note that I've deleted all the slack constraints ($c_t \geq 0, t = 0, 1, \dots, T$, and $k_t \geq 0, t = 1, 2, \dots, T$) in the above problem, which simplifies your derivation. (Even if I include these constraints in the above, the multipliers for these constraints would be zero because they do not bind.)

2. (50 points) Consider an economy with two price-taking firms. Firm 1 has a production technology $f(k_1, n_1) = k_1^{1/3} n_1^{2/3} + k_1$, while firm 2 has $g(k_2, n_2) = k_2^{2/3} n_2^{1/3} + n_2$. Suppose the total supply of capital is 5 ($k_1 + k_2 = 5$), and the total supply of labor is 3 ($n_1 + n_2 = 3$).

- (a) Formulate the equilibrium condition as a system of non-linear equations. (15 points)
- (b) Use newton method to find the equilibrium allocations and price levels. (35 points) (The system of equations should be modified before Newton's method is applicable.)